Executive Summary: Do Early Bookings Cancel More at the City Hotel?

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Business question

This analysis is for the Revenue Manager of the City Hotel. The decision is whether the hotel should make its cancellation rules stricter for bookings made more than 90 days before arrival, for example by asking for a small deposit or by allowing free cancellation only until 30 days before arrival.

My research question was: do City Hotel bookings made more than 90 days ahead get cancelled more often than bookings made 90 days or less ahead, and is the difference big enough to matter? Every cancelled booking is a room the hotel held but did not sell. The City Hotel cancels about 42% of its bookings, compared with 28% at the Resort Hotel, so cancellations are a real problem for its revenue. Because a stricter rule could also push some guests away, I decided before the analysis that a gap of at least 10 percentage points would be big enough to act on.

Data

I used the Hotel Booking Demand dataset, where each row is one booking at either the City Hotel or the Resort Hotel. There were 119,390 bookings. I checked missing values, data types, duplicates, outliers and values that do not make sense. I removed 184 bookings: 180 with no guests, one with a negative price, one at $5,400 per night (the next highest was $510) and two with an undefined market segment. That left 119,206 bookings, of which 79,160 were at the City Hotel. About 27% of the rows were exact duplicates. Because there is no booking ID, I could not tell if they were errors or real group bookings, so I kept them and checked my result without them. I split the bookings into Early (more than 90 days before arrival) and Late (90 days or less). The 90-day cut-off is a practical business threshold I chose in advance, not a natural break in the data.

Most relevant findings

- 44% of City Hotel bookings are made more than 90 days ahead, so early bookings are a big part of the business.
- 56% of early bookings were cancelled, compared with 31% of late bookings. The price per night was almost the same in both groups ($102 vs $108), so the difference is about cancellations, not price.
- The further ahead people book, the more they cancel. At the City Hotel it goes from 12% for bookings made within a week to 71% for bookings made more than a year ahead (Figure 1). The Resort Hotel shows the same pattern, but weaker.

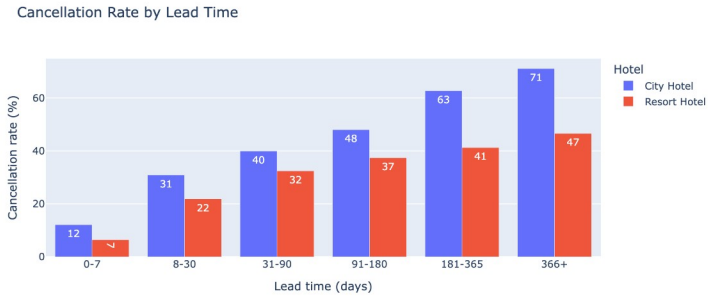


Figure 1: Cancellation rate by lead time for both hotels (n = 119,206 bookings; City Hotel is the focus).

Statistical result

I used a z-test for two proportions, because the outcome is yes/no (cancelled or not) and I am comparing two large, separate groups. My hypotheses were H_0: early and late bookings have the same cancellation rate, and H_1: early bookings have a higher cancellation rate ($\alpha = 0.05$).

City Hotel	Early (> 90 days)	Late (≤ 90 days)	Difference
Bookings	35,143	44,017	
Cancelled	55.8%	30.6%	25.1 points
95% confidence interval	55.2% – 56.3%	30.2% – 31.1%	24.5 – 25.8 points

The z-statistic was 71.2 and the p-value was below 0.001, so I rejected H_0. Early bookings cancel 25.1 percentage points more often than late bookings, with a 95% confidence interval of 24.5 to 25.8 points. That is almost twice as often, well above my 10-point threshold. I double-checked the calculation with a chi-square test on the same data, and it matched.

If early bookings cancelled at the same rate as late bookings, there would have been about 8,800 fewer cancellations. At the average room revenue of a cancelled early booking ($315), that is an illustrative gross exposure of about $2.8 million if no rooms were resold, $1.4 million if half were, and $0.7 million if three quarters were. It is not a measured loss, and not the revenue a new policy would recover.

What supports and what challenges the result

Supporting the result, the same pattern appears at the Resort Hotel (39% of early bookings cancelled vs 20% of late ones), so it is not something that only happens at one hotel.

Challenging the result, early bookings include many more group bookings (27% vs 10%) and offline travel agent bookings (27% vs 16%), and these segments cancel a lot no matter when they book. So part of the gap comes from who books early. Because identical rows may not be independent bookings, I repeated the z-test without them: the gap shrank to 14.0 points (95% CI: 13.2 to 14.9, $z = 33.3$), but even the bottom of that interval is above my threshold. However, early bookings still cancel more inside every main segment. When I repeated the z-test only for Online TA, the largest segment with almost no group contracts, early bookings still cancelled 46% vs 32% ($z = 27.0$, $p < 0.001$). The gap is smaller, 13.7 points (95% CI: 12.7 to 14.7), but still above my threshold.

My interpretation: book now, cancel later

The data shows that early Online TA bookings cancel more, but not why. My hypothesis is "book now, cancel later". Online travel agencies push free cancellation, so a guest can grab a room months ahead as a free placeholder, keep looking, and cancel if they find something better. The data cannot show intent, but two checks are consistent with it. First, 99.9% of Online TA bookings have no deposit, so cancelling costs nothing. Second, among early Online TA bookings, guests who made no special request cancelled 66% of the time (5,754 bookings), compared with 33% for guests who made at least one (8,553 bookings) (Figure 2). A guest who asks for a quiet room or a crib is planning a real trip; a guest who never touches the booking looks more like a placeholder.



Figure 2: Early Online TA bookings at the City Hotel, cancellation rate by special request (n = 14,307).

This is still an association, so special requests are a signal to test, not a proven predictor. If cancellation dates showed that most cancellations happen close to arrival, it would point more to "plans changed" than to "book now, cancel later".

One conclusion I cannot make is about deposits. Almost 100% of non-refundable bookings were cancelled, but this does not mean deposits do not work or that deposits cause cancellations. 97% of them come from Groups and offline travel agents, with a median lead time of 188 days, which are the bookings that cancel the most anyway. The data cannot show how guests would react to a new deposit rule. I also cannot say that booking early causes people to cancel, only that the two are strongly related.

Recommendation

1. Shorten the free-cancellation window for Online TA bookings made more than 90 days ahead, for example free cancellation only until 30 days before arrival. I prefer this to a deposit because it targets free placeholders directly without scaring away real guests. Run it on some arrival dates for 8 to 12 weeks and compare with dates that keep the normal rule. I would call it a success if the early Online TA cancellation rate falls by at least 5 points (from about 46%) while early Online TA bookings fall by no more than 5%.
2. Use special requests as an early warning: send early Online TA guests with no special request a reconfirmation email about 45 days before arrival, and treat them as higher risk when deciding how much to overbook.
3. Manage group and offline agent bookings through their contracts (for example release dates), since their cancellations are not only about lead time.

My analysis shows an association, not that these changes will work, and the 30-day and 45-day figures are proposals to test; that is why I recommend a pilot. The recommendation is supported by the 25-point gap, the steady increase with lead time, the 14-point gaps without duplicates and inside Online TA, and the 66% vs 33% difference by special request. The main risk is that a shorter window could push some guests to a competitor that still offers free cancellation, and this data cannot tell me how many. To decide better, the hotel needs the test results, the date each booking was cancelled, and whether cancelled guests rebooked at a lower price.

Principal limitation

The biggest limitation is that this is observational data from only two hotels, with no booking ID, no cancellation date and no information on whether rooms were resold. Some bookings, such as groups, are probably not independent, so the confidence intervals may be too narrow. The results show a strong association, but they cannot prove that booking early causes cancellations or predict how guests will react to a new policy.